

MUHAMMAD ABDULLAH BUTT

Full Stack Engineer & AI Researcher | Machine Learning, Computer Vision & LLM/RAG Systems

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PROFESSIONAL SUMMARY

Full Stack Software Engineer and AI/ML Researcher with 3+ years of professional experience designing, building, and deploying scalable web applications, mobile applications, and production machine learning systems. Published author of MAPF-Lite, a peer-reviewed multimodal deepfake detection framework achieving 99.34% accuracy at 0.566M parameters, accepted at FLINS-ISKE 2026 and included in a Springer Nature proceedings volume. Shipped 7+ live production products spanning healthcare, fintech, livestock management, fashion e-commerce, image processing, and SaaS. Deep expertise in Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Natural Language Processing (NLP), Large Language Models (LLM), Retrieval-Augmented Generation (RAG) pipelines, agentic AI workflows, React Native mobile development, and full-stack web development using Next.js, Node.js, and Python. Skilled in API development, cloud deployment, CI/CD automation, and Agile software delivery. Certified Oracle Generative AI Professional and IBM Agentic AI Expert.

RESEARCH PUBLICATIONS

MAPF-Lite: A Parameter-Efficient Multimodal Framework for High-Fidelity Deepfake Detection

FLINS-ISKE Journal, 2026 (Published) — Springer Nature Proceedings Volume — Sole architect and lead researcher

- 99.34% accuracy and 99.84% AUC on FakeAVCeleb; 99.40% cross-dataset accuracy on DeepfakeTIMIT (zero-shot)
- Reduced trainable parameters by 99.6% to 0.566M via frozen CLIP/Whisper backbones with SRM and DCT analysis
- 4.7x real-time inference at 1.1 GB peak memory; edge and mobile deployable
- Explainable AI (XAI) modules with visual heatmaps and frame-level suspicion scores for forensic transparency

Deepfake Detection in the Multimodal Era: A Systematic Review

Review Paper (In Preparation) — Lead author

- Systematic literature review surveying multimodal deepfake detection techniques, parameter-efficient architectures, and benchmark datasets, extending MAPF-Lite findings

WORK EXPERIENCE

Senior Full-Stack AI Engineer

Mar 2026 - Present

Huzzle.com | London, UK (Remote)

- Building AI-driven recruitment infrastructure integrating LLM pipelines with full-stack product features for a live, high-traffic B2C platform

Full Stack Engineer

Jan 2023 - Feb 2026

MAQ Enterprises Ltd. | Rawalpindi, Pakistan

- Engineered production web applications with Next.js and Node.js; reduced page load time by 40% via server-side rendering (SSR) optimization and code splitting
- Designed RESTful APIs and optimized PostgreSQL and MongoDB database schemas, improving data retrieval speed by 35% on critical endpoints
- Implemented Jenkins and Docker CI/CD pipelines, cutting release cycle time by 30%
- Automated end-to-end software testing with Cypress and Jest, reducing production bug incidence by 25%
- Mentored 3+ junior developers through code reviews; provisioned infrastructure-as-code (IaC) using Terraform

AI Researcher

2024 - 2026

GIFT University | Gujranwala, Pakistan

- Designed MAPF-Lite multimodal deep learning architecture (CLIP + Whisper + ViT) for deepfake detection, published in peer-reviewed journal and Springer Nature proceedings

- Reduced trainable parameters by 99.6% while maintaining state-of-the-art accuracy through parameter-efficient fine-tuning (PEFT)
- Achieved 30% reduction in detection latency by developing lightweight attention mechanisms for edge AI deployment

Full-Stack Developer

2025 - May 2026

Estabraq (Fashion Brand) | Pakistan

- Architected live fashion e-commerce platform (estabraq.pk) with Next.js App Router, Prisma ORM, and Supabase; achieved sub-2-second load times in production
- Integrated GraphQL APIs, Redis caching, and CDN image optimization, reducing server latency by 30%
- Automated deployment with Docker CI/CD, cutting release cycles by 25%

KEY PROJECTS

DeepShield — MAPF-Lite-Based Deepfake Detection Web Platform (Launching Soon)

Visit: [deepshield.video](#)

Next.js - FastAPI - PyTorch - CLIP - Whisper - Computer Vision - Deep Learning

- Building a production web platform applying the published MAPF-Lite multimodal architecture to real-time deepfake detection for uploaded video and audio content, launching within days
- Productionizing the research-grade CLIP/Whisper-based model into a scalable web service with an accessible upload-and-analyze workflow
- Bringing 99%+ accuracy detection and XAI-based forensic heatmaps from the published research into a consumer- and enterprise-facing tool

ImageStudio — Privacy-First Online Image Toolkit

Visit: [imagestudio.app](#)

Next.js - Image Processing - Headless Browser Rendering - Cloud Storage APIs

- Built a privacy-first, account-free image utility platform with automatic file deletion after 2 hours, offering compression, resizing, cropping/rotation, and format conversion (HEIC, RAW, PSD, TIFF, JPG, PNG, WEBP, animated GIF)
- Developed creative tools including a canvas-based photo editor (text, stickers, frames, filters), a meme generator, and configurable watermarking with tiling, opacity, and blend modes
- Implemented an HTML-to-image renderer using a headless browser for full-page screenshots across mobile and desktop viewports
- Built batch processing supporting 20 simultaneous image uploads with ZIP export, plus cloud import integrations (Google Drive, Dropbox, URL)

Pointr — Livestock Management Platform (Web + Mobile)

Visit: [livestock.pointrsolutions.co.za](#)

Expo - React Native - Django - JWT - Android - iOS - EAS

- Built and published a full-stack livestock management ecosystem: web dashboard + Expo/React Native mobile app (Android + iOS) with real-time sync via Django backend and JWT authentication
- Engineered offline-first two-phase architecture: data captured in the field without connectivity, uploaded on mobile data — zero data loss across 100% of sync cycles in production
- Integrated XR5000 weigh-scale via Wi-Fi, parsing ADI XML sessions (EID, weight, timestamp, health notes); UUID-keyed uploads eliminate duplicate weigh-ins on retry
- Wrote back clean animal data (EID, breed, weight history) to scale indicator, replacing bare tag numbers with real farm records
- Isolated all device logic behind a single adapter + mock server; ~90% of development done without hardware, reducing debugging cost significantly
- Released via EAS: signed .aab for Google Play with OTA updates for JS-only fixes, bypassing full store review cycles

Estabraq — Fashion E-Commerce Platform

Visit: [estabraq.pk](#)

Next.js - TypeScript - Supabase - Prisma ORM - GraphQL - Redis

- Live production platform with sub-2-second load times, real-time inventory, and secure authentication
- GraphQL APIs + Redis caching reduced client-server latency by 30%; CDN image optimization cut bounce rates measurably
- React Native mobile companion app in progress for omnichannel experience

MyPDFMate — PDF SaaS

Visit: mypdfmate.com

Next.js - FastAPI - AI Processing

- Live production SaaS platform for PDF processing and AI-powered document workflows

Khushi Motors — Automotive Platform

Visit: khushimotors.com

Next.js - Full Stack Web Development

- Live automotive marketplace with full product catalogue, filtering, and inquiry management

LingumedAI — Clinical AI for Nurses

React - FastAPI - LangChain - RAG - pgvector - OpenAI API

- AI clinical assistant giving nurses real-time access to diagnostic guidance and drug interaction checks, reducing physician dependency in critical moments
- Domain-specific RAG pipeline ingesting medical literature and clinical guidelines into pgvector; hallucination-resistant responses with explicit uncertainty escalation
- Role-based access control and audit logging for healthcare data governance compliance

ForexAI — Mobile Trading & Prediction App

React Native - XGBoost - Kronos AI - Python

- Forex education app with AI market prediction; fine-tuned XGBoost + Kronos AI ensemble on historical price data
- In-app learning curriculum paired with live prediction signals for retail traders

TECHNICAL SKILLS

Category	Skills
AI & LLM	LangChain, LlamaIndex, OpenAI API, Anthropic Claude, RAG Pipelines, Pinecone, pgvector, Agentic Workflows, Prompt Engineering
Machine Learning & Fintech AI	XGBoost, Decision Trees, Kronos AI, Model Fine-Tuning, Ensemble Modeling, Time-Series Forecasting
Computer Vision & Deep Learning	PyTorch, Vision Transformers (ViT), CLIP, Whisper, PEFT, Model Quantization, Edge AI Deployment
Frontend Development	Next.js, React, React Native, Expo, TypeScript, JavaScript, Tailwind CSS, GraphQL
Backend Development	FastAPI, Node.js, Django, Prisma ORM, RESTful API Design, Redis
Databases	PostgreSQL, MongoDB, Supabase
DevOps & Cloud	Docker, Jenkins, Terraform, CI/CD, EAS, Vercel, Railway, Oracle Cloud, AWS
Testing & Quality	Cypress, Jest, End-to-End Testing, Unit Testing
Methodologies	Agile, Software Development Life Cycle (SDLC), Version Control (Git/GitHub), Code Review, Cross-Functional Collaboration

EDUCATION

Bachelor of Science in Computer Science

Sep 2022 - Mar 2026

GIFT University, Gujranwala, Pakistan

- Thesis: MAPF-Lite — published at FLINS-ISKE 2026 and included in Springer Nature proceedings volume

CERTIFICATIONS

- OCI Certified Generative AI Professional - Oracle, 2025
- OCI Certified AI Foundations Associate - Oracle, 2025
- Agentic AI Expert - IBM
- Build RAG Applications - IBM, 2026
- NLP and Computer Vision Specialist - IBM
- Generative AI: Language Modeling with Transformers - IBM
- Advanced React - Meta
- Machine Learning Using Python - IBM